



DM,
Autumn, 2026



Correlation & Association Rules

Faculty of DS & AI
Autumn semester, 2026

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Content

- **Correlation**
- Association Rules
- Tools

Correlation



DESCRIBE

Goal: Summarize and visualize data.

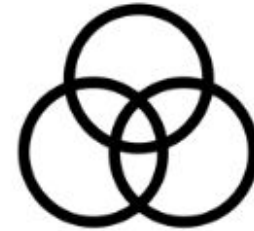
Key Question: What is the data saying?



RELATE

Goal: Identify correlations and associations.

Key Question: What usually goes together?



GROUP

Goal: Cluster objects with shared, natural traits.

Key Question: How many groups can these objects be divided into?



PREDICT

Goal: Utilize classification and regression.

Key Question: What will happen next?

Correlation

Definition

- Sarah is a regional sales manager for a company that supplies residential heating oil.
- Prices fluctuate, and orders are inconsistent.
- Sarah wants to understand: what factors are related to **heating oil consumption**?



Correlation

Definition

- Home insulation.
- Outdoor temperature.
- Number of occupants.
- Average age
- House size.

- **Correlation** is a statistical measure of **how strong the association** between two variables is.
- Pre-visualization using **scatter plot**: each point is an observation;
- If the points tend to lie along an upward/downward line → there is a connection.

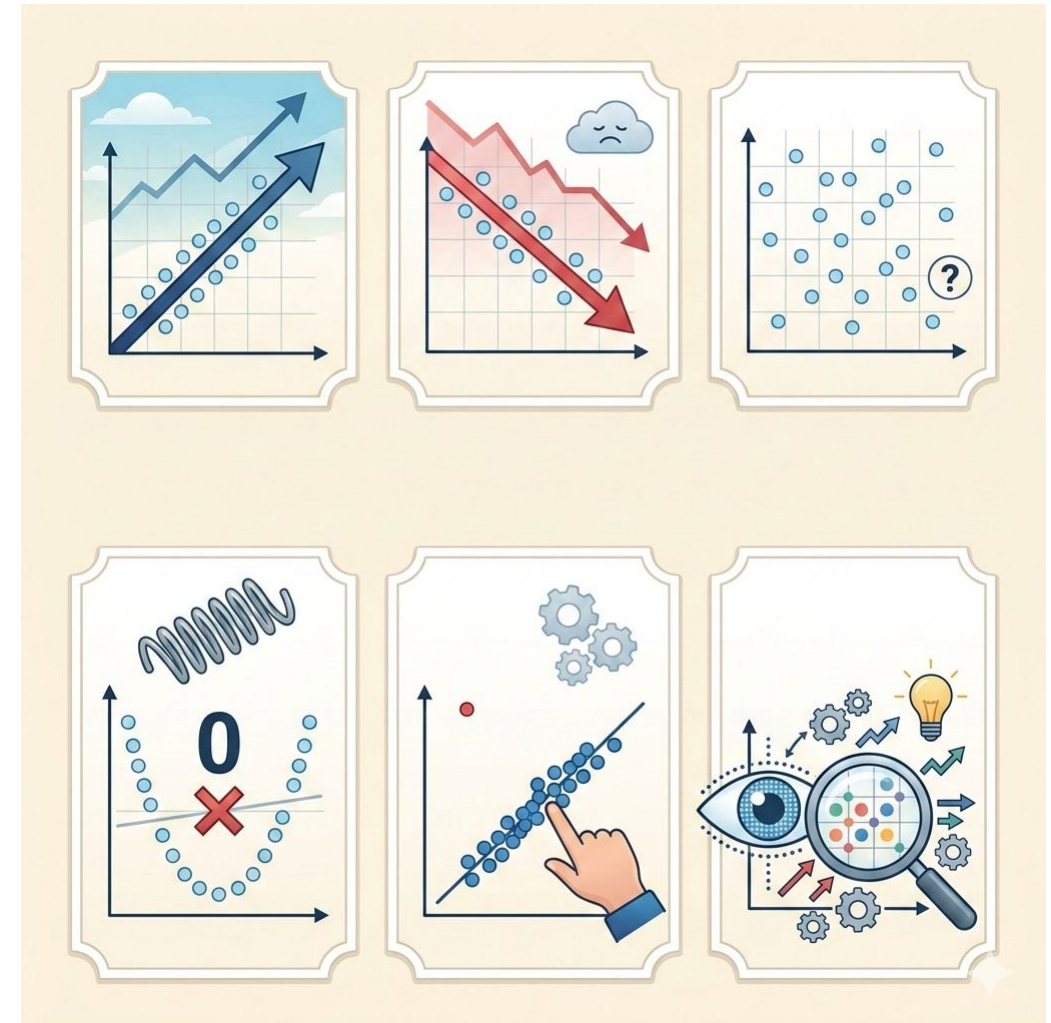


Correlation

Definition

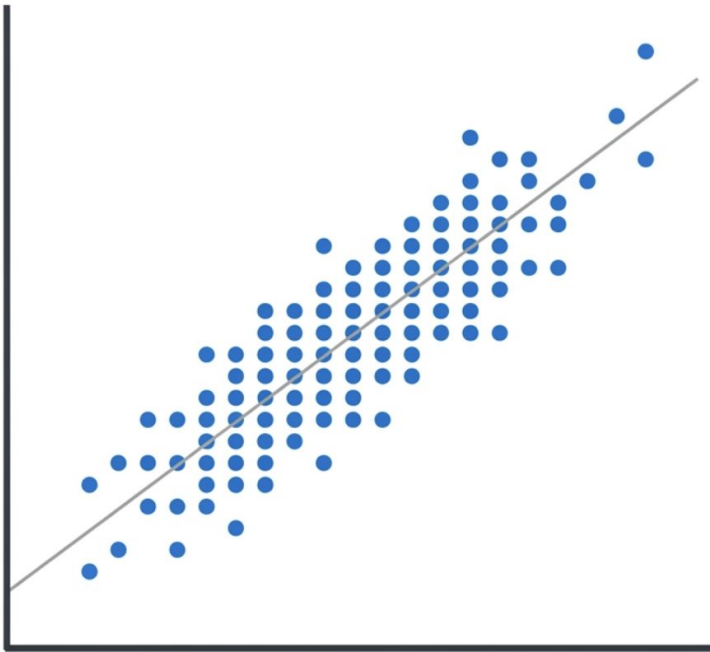
Scatter plot help:

- **See the direction and strength with your eyes:** points line up to the right → positive; going down → negative; scattered in clusters → uncorrelated.
- **Detecting nonlinear relationships:** two variables with a curved relationship (eg U-shaped) can have $r \approx 0$ even though they are related.
- **Outlier detection:** an outlier can skew the coefficients.
- Verify the validity of the **correlation coefficient**.

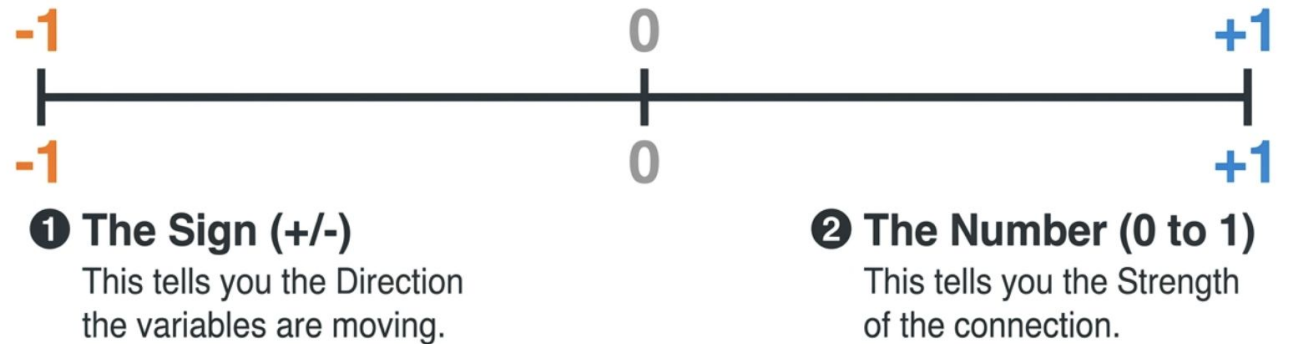


Correlation

Definition



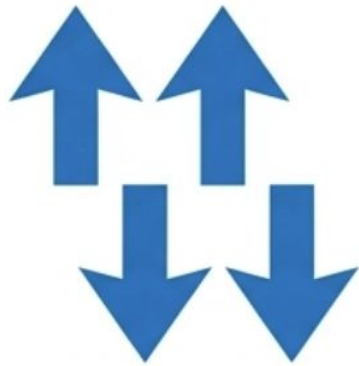
The coefficient reveals both direction and strength.



Correlation

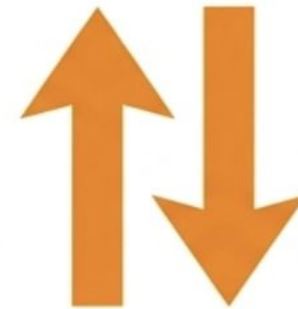
Definition

Positive correlation means moving together, even if both are dropping.



Positive (+)

Variables move in the same direction.
When one goes up, the other goes up.
When one goes down, the other goes down.

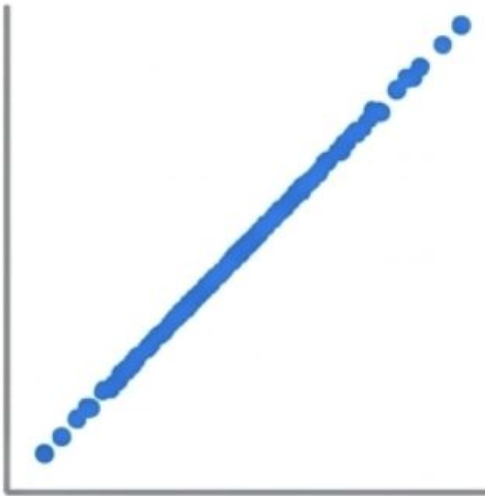


Negative (-)

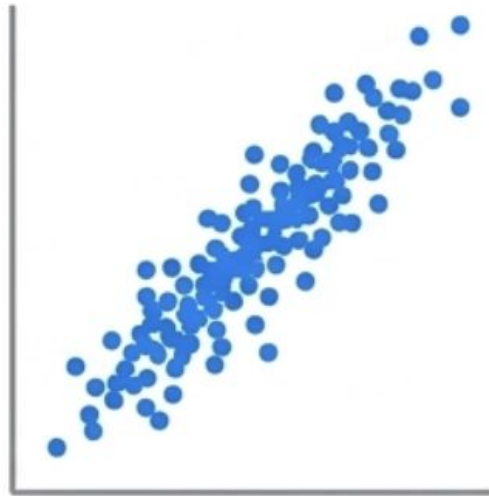
Variables move in opposite directions.
When one goes up, the other goes down.

Correlation

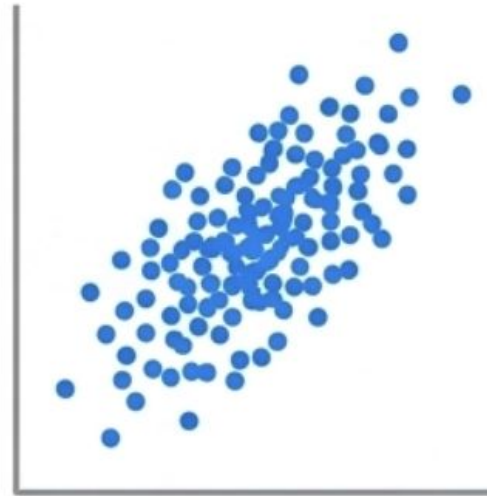
Definition



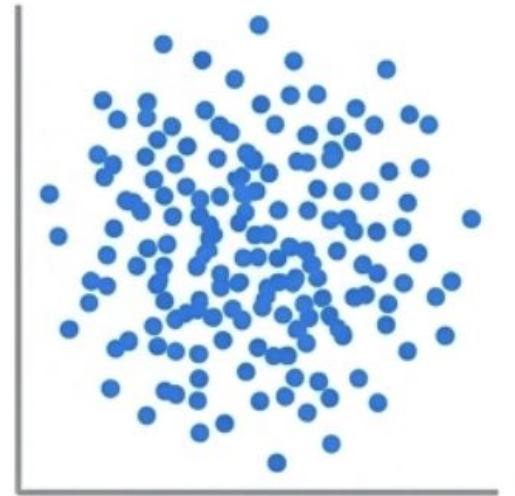
0.8 to 1.0
Very Strong



0.6 to 0.8
Strong



0.4 to 0.6
Correlated



0.0 to 0.4
Weak / None

Correlation

Definition

Pearson Correlation Coefficient — General Formula

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} \quad (1)$$

- ****r**** — the **Pearson correlation coefficient**. It measures the strength and direction of the *linear* relationship between two numeric variables. Its value always lies between **-1 and +1**.
- x_i, y_i — the values of the two variables for the ***i*-th observation** (e.g., the Insulation and Heating Oil values of house *i*).
- n — the **number of observations** (the number of paired data points; in the example, $n = 4$ houses).
- $\bar{x}, \bar{y}$ — the **means** (averages) of variable x and variable y :

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i, \quad \bar{y} = \frac{1}{n} \sum_{i=1}^n y_i$$

Correlation

Definition

deviations of each value from its own mean. They tell us whether a value is *above* (positive) or *below* (negative) the average of its variable.

Pearson Correlation Coefficient — General Formula

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} \quad (1)$$

Correlation

Definition

the numerator: the sum of the products of the paired deviations.

Pearson Correlation Coefficient — General Formula

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} \quad (1)$$

- If both deviations share the same sign (both above or both below their means), the product is **positive** → the variables move **together**.
- If they have opposite signs, the product is **negative** → the variables move in **opposite directions**.
- The **sign of this whole sum decides the direction** of the correlation (positive or negative). *(This quantity, divided by n , is the **covariance** of X and Y .)*

Correlation

Definition

The denominator: a measure of the spread (dispersion) of each variable around its mean.

Pearson Correlation Coefficient — General Formula

$$r = \frac{\sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum_{i=1}^n (x_i - \bar{x})^2} \sqrt{\sum_{i=1}^n (y_i - \bar{y})^2}} \quad (1)$$

Correlation

Example

The target variable is **Heating_Oil** — the annual consumption of heating oil (in units).

- Apply func. (1) to calculate r for each variable along with target var.
- Use scatter plot to visualize the correlation between each variable to the target.

Home	Insulation	Avg_Age	Num_Occupants	Home_Size	Heating_Oil
H1	3	20	3	4	180
H2	4	68	2	4	230
H3	5	25	5	4	260
H4	6	63	4	4	300
H5	6	30	3	4	330
H6	7	58	5	4	360
H7	8	35	2	4	400
H8	9	52	4	4	440
H9	10	40	3	4	470
H10	11	45	4	4	520
H11	12	47	12	8	560

Correlation

Example

Home	Insulation	Avg_Age	Num_Occupants	Home_Size	Heating_Oil
H1	3	20	3	4	180
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Bước 1 — Trung bình: $\bar{x} = 81 / 11 = 7.3636$; $\bar{y} = 4050 / 11 = 368.1818$

Bước 2 — Lập bảng độ lệch và tích cho từng ngôi nhà:

Nhà	x (Insul.)	y (Oil)	$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$	$(x - \bar{x})^2$
H1	3	180	-4.364	-188.18	+821.16	19.041
H2	4	230	-3.364	-138.18	+464.79	11.314
H3	5	260	-2.364	-108.18	+255.70	5.587
H4	6	300	-1.364	-68.18	+92.98	1.860
H5	6	330	-1.364	-38.18	+52.07	1.860
H6	7	360	-0.364	-8.18	+2.98	0.132
H7	8	400	+0.636	+31.82	+20.25	0.405
H8	9	440	+1.636	+71.82	+117.52	2.678
H9	10	470	+2.636	+101.82	+268.43	6.950
H10	11	520	+3.636	+151.82	+552.07	13.223
H11	12	560	+4.636	+191.82	+889.34	21.496
TỔNG					3537.27	84.545

Bước 3 — Tử số = tổng cột $(x - \bar{x})(y - \bar{y}) = 3537.27$

Bước 4 — $\Sigma(x - \bar{x})^2 = 84.545$; $\Sigma(y - \bar{y})^2 = 148763.6$ (đã tính ở mục 3)

Bước 5 — Mẫu số = $\sqrt{84.545} \times \sqrt{148763.6} = 9.195 \times 385.70 = 3546.45$

Bước 6 — $r = 3537.27 / 3546.45 = +0.9974$

Correlation

Example

Bước 1 — Trung bình: $\bar{x} = 81 / 11 = 7.3636$; $\bar{y} = 4050 / 11 = 368.1818$

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H11	12	560	+4.636	+191.82	+889.34	21.496
TỔNG					3537.27	84.545



Trung bình target $\bar{y}$ (Heating_Oil) = 368.18. Vì cùng target nên $\Sigma(y_i - \bar{y})^2 = 148763.6$ chung cho mọi biến.

Bước 3 — Tử số = tổng cột $(x - \bar{x})(y - \bar{y}) = 3537.27$

Bước 4 — $\Sigma(x - \bar{x})^2 = 84.545$; $\Sigma(y - \bar{y})^2 = 148763.6$ (đã tính ở mục 3)

Bước 5 — Mẫu số = $\sqrt{84.545} \times \sqrt{148763.6} = 9.195 \times 385.70 = 3546.45$

Bước 6 — $r = 3537.27 / 3546.45 = +0.9974$

Biến (x)	$\bar{x}$	Tử số $\Sigma(x - \bar{x})(y - \bar{y})$	$\Sigma(x - \bar{x})^2$	$v(S_{xx} \cdot S_{yy})$	r
Insulation	7.364	3537.3	84.55	3546.4	+0.9974
Avg_Age	43.909	2988.2	2436.91	19040.0	+0.1569
Num_Occupants	4.273	1755.5	76.18	3366.5	+0.5215
Home_Size	4.364	767.3	14.55	1471.0	+0.5216

Correlation

Example

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt

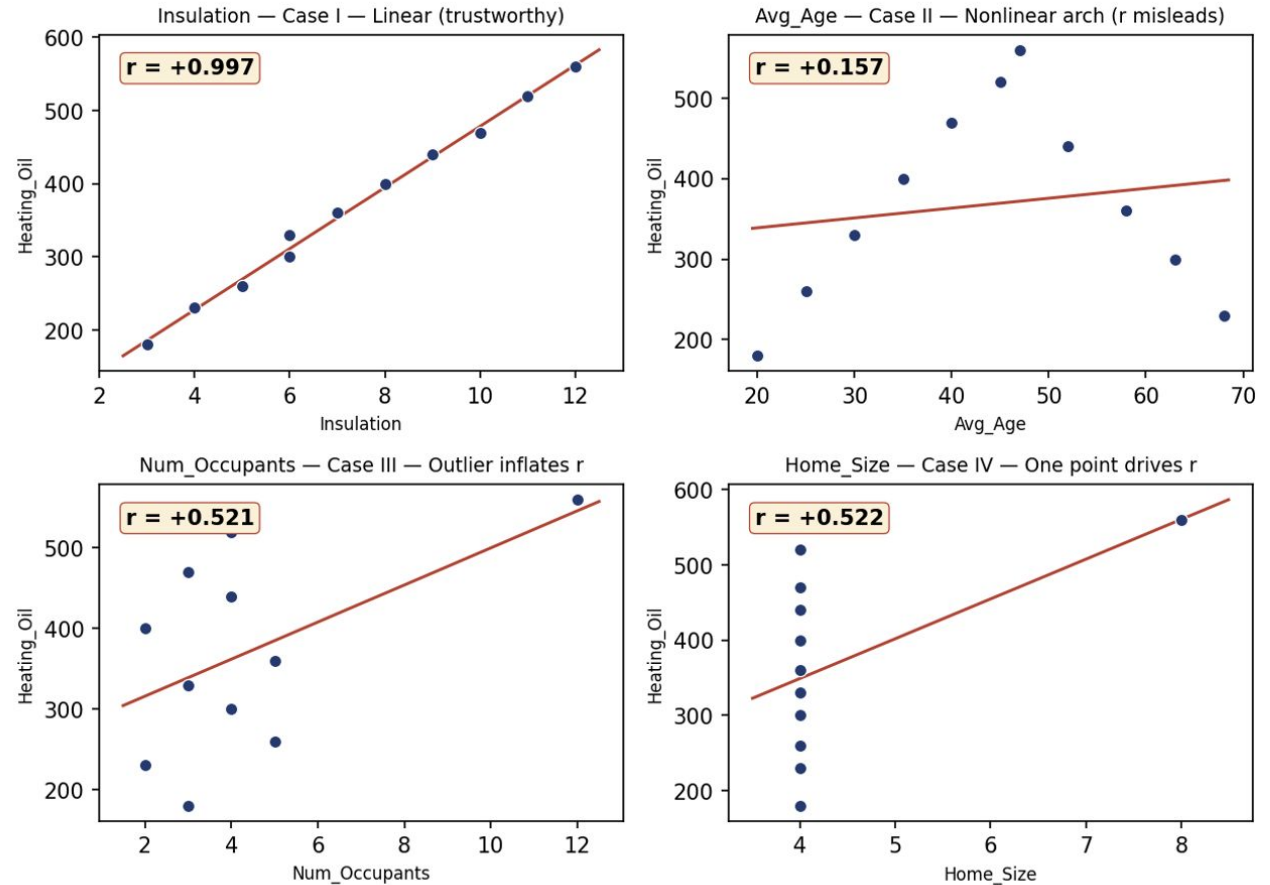
df = pd.read_csv("heating_oil_dataset.csv")
oil = df["Heating_Oil"]

cols = ["Insulation", "Avg_Age", "Num_Occupants", "Home_Size"]
fig, axes = plt.subplots(2, 2, figsize=(9, 7))

for ax, col in zip(axes.flat, cols):
    x = df[col]
    r = np.corrcoef(x, oil)[0, 1]          # he so tuong quan r
    ax.scatter(x, oil)                    # ve cac diem
    b, a = np.polyfit(x, oil, 1)          # duong xu huong (hoi quy)
    ax.plot(x, a + b * x, color="red")
    ax.set_title(f"{col} (r = {r:.3f})")
    ax.set_xlabel(col)
    ax.set_ylabel("Heating_Oil")

plt.tight_layout()
plt.savefig("heating_oil_scatter.png", dpi=150)
plt.show()
```

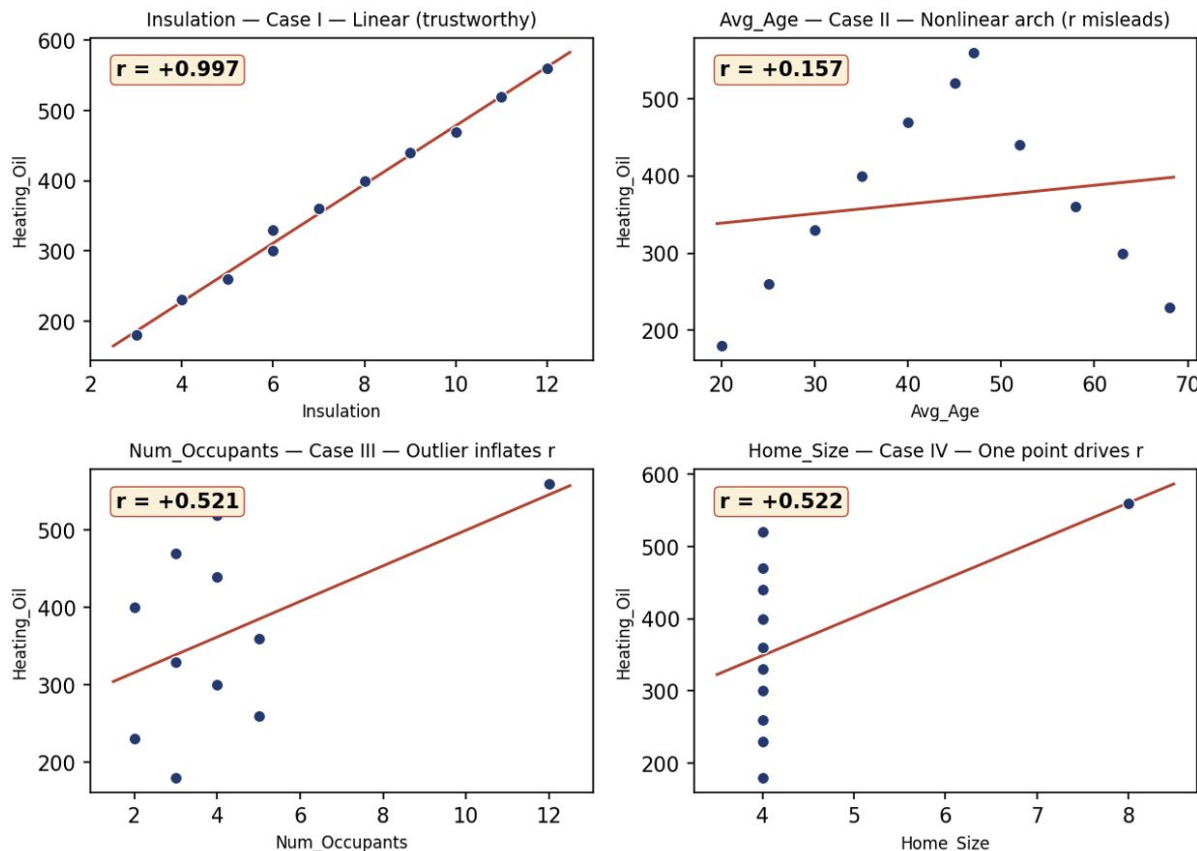
Heating Oil vs 4 factors — same target, four different patterns



Correlation

Example

Heating Oil vs 4 factors — same target, four different patterns



- Insulation: $r = +0.997$ — very strong, positive.
-> The **points** stick close to the upward **straight line**. This is a clear and reliable **LINEAR relationship**: the more insulated the house, the more oil it consumes. r reflects reality.

- Avg_Age
-> $r = +0.157$ — close to 0, seemingly 'irrelevant'. BUT the scatter plot shows an **inverted-U** shape: middle-aged people use the most oil, and very young and very old people use the least. The relationship is **STRONG** but **NONLINEAR**, so linear $r \approx 0$ hides it

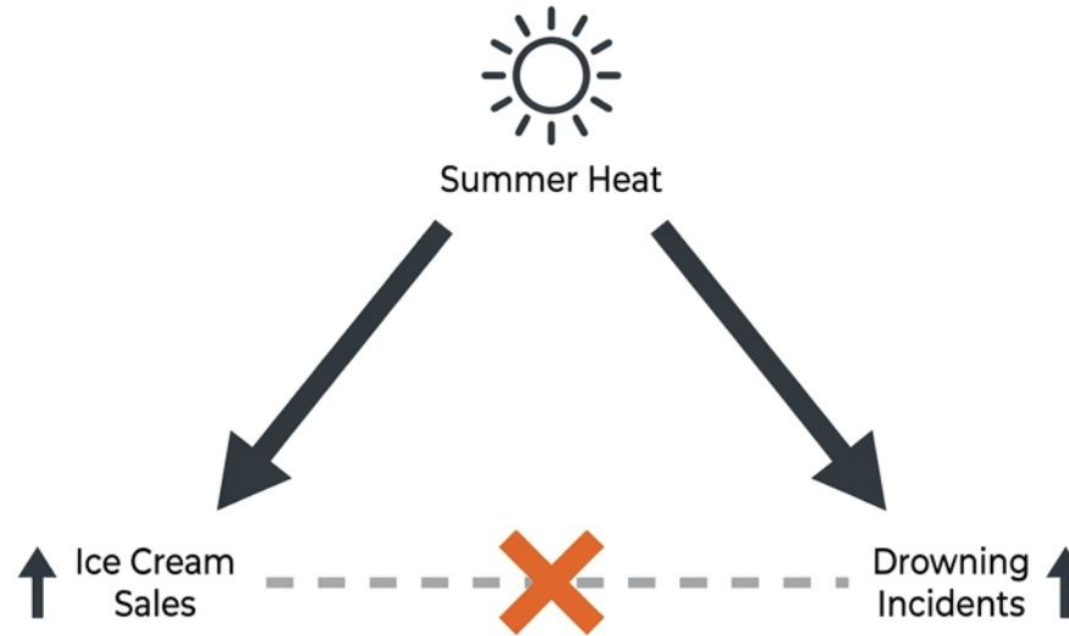
- Num_Occupants
-> $r = +0.521$ — average, looks worth using. But this r is mainly pulled up by **ONE** outlier (house of 12 people, uses 560 oil). Removing that point, r is only ≈ 0.12 .

- Home_Size
-> $r = +0.522$ — average, but meaningless. Nearly every house has Home_Size = 4 (one vertical column), only **ONE** house = 8. The entire 'variation' of x is at a single point $\rightarrow r$ is determined by that point.

Correlation

Correlation vs. Causation

Two variables moving together does not equal causation.



This is the golden rule of analytics. A high correlation coefficient might just be a coincidence, or it might be driven by a hidden 'Third Variable.'

The BA Rule: Always ask 'Why?' before assuming causation.

Content

- Correlation
- **Association Rules**
- Tools

Association Rules

Definition

- **Association Rules** finds items that frequently appear together in transactions — which is market-basket analysis. The results are rules of the form:

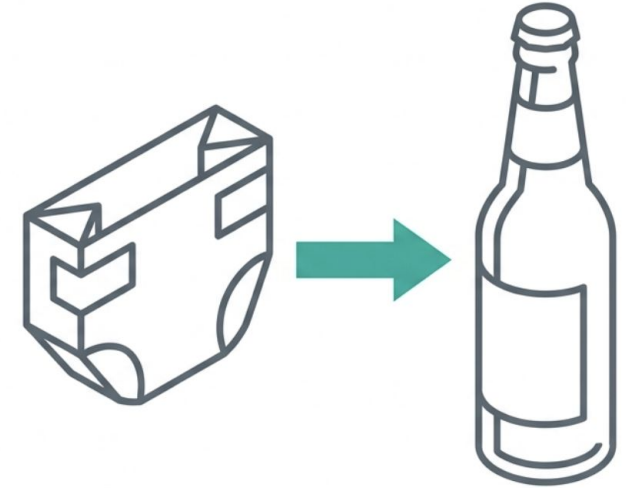
$\Rightarrow A \rightarrow B$ ("if customers buy A, they usually buy B")

- **3 Counting indices:**
 - **Support:** How often does this rule come up?
 - **Confidence:** Among those who purchased product A, what percentage also purchased product B?
 - **Lift:** Is group buying really more than just random chance?

The Hidden Patterns in the Basket

What happens when fathers make a late-night diaper run? They buy beer.

This is a rule. $A \rightarrow B$ (If a customer buys A, they often buy B). Supermarkets use this exact logic to rearrange shelves, build combos, and design promotions.



Association Rules

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$\Rightarrow A \rightarrow B$ ("if customers buy A, they usually buy B")

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 - **Lift:** Is group buying really more than just random chance?

$$\text{Support}(A \rightarrow B) = \frac{\text{count}(A \cap B)}{N} = P(A \cap B)$$

The proportion of transactions containing both A and B; indicates how frequently the rule occurs.

$$\text{Confidence}(A \rightarrow B) = \frac{\text{count}(A \cap B)}{\text{count}(A)} = \frac{\text{Support}(A \cap B)}{\text{Support}(A)} = P$$

Among transactions involving A, what is the proportion that also includes B? This indicates the reliability of the rule.

$$\text{Lift}(A \rightarrow B) = \frac{\text{Confidence}(A \rightarrow B)}{\text{Support}(B)} = \frac{\text{Support}(A \cap B)}{\text{Support}(A) \cdot \text{Support}(B)} = \frac{P(A \cap B)}{P(A) P(B)}$$

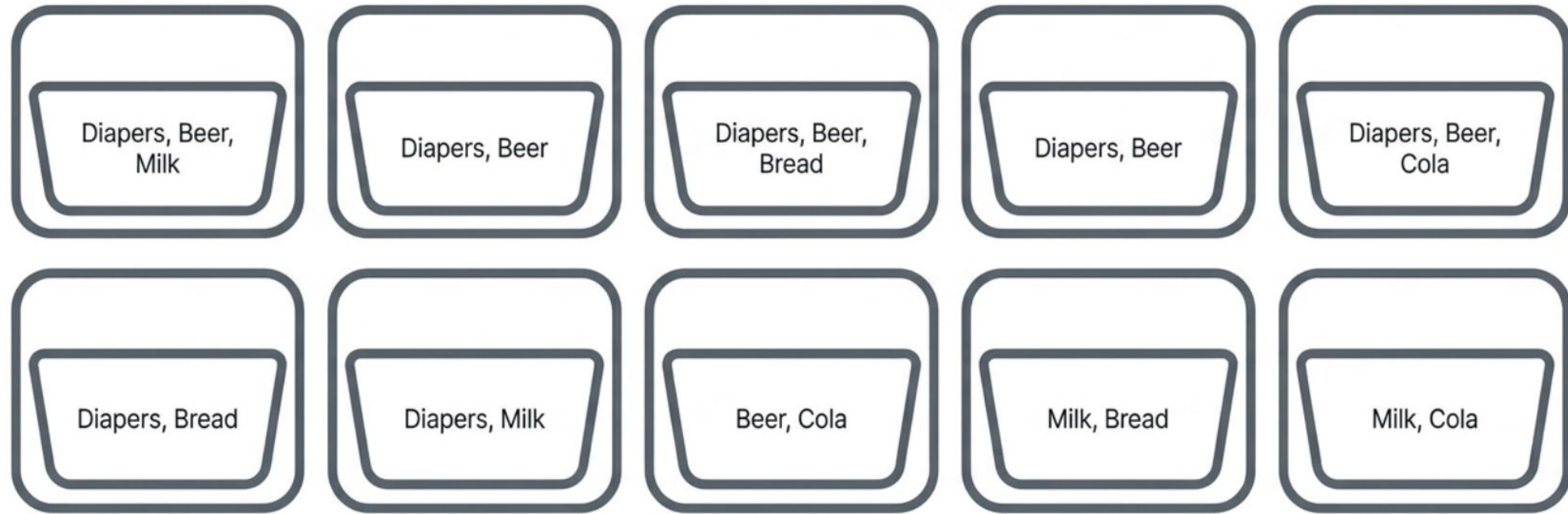
Compare the "A and B together" level to the random level. Read:

- *Lift > 1: go together more than by chance → positive contact, worth using.*
- *Lift = 1: independent → meaningless law.*
- *Lift < 1: come together less than by chance → exclude each other.*

Association Rules

Example

To decode the rules, we must first count. Our **universe** consists of **Total Transactions (N) = 10**.



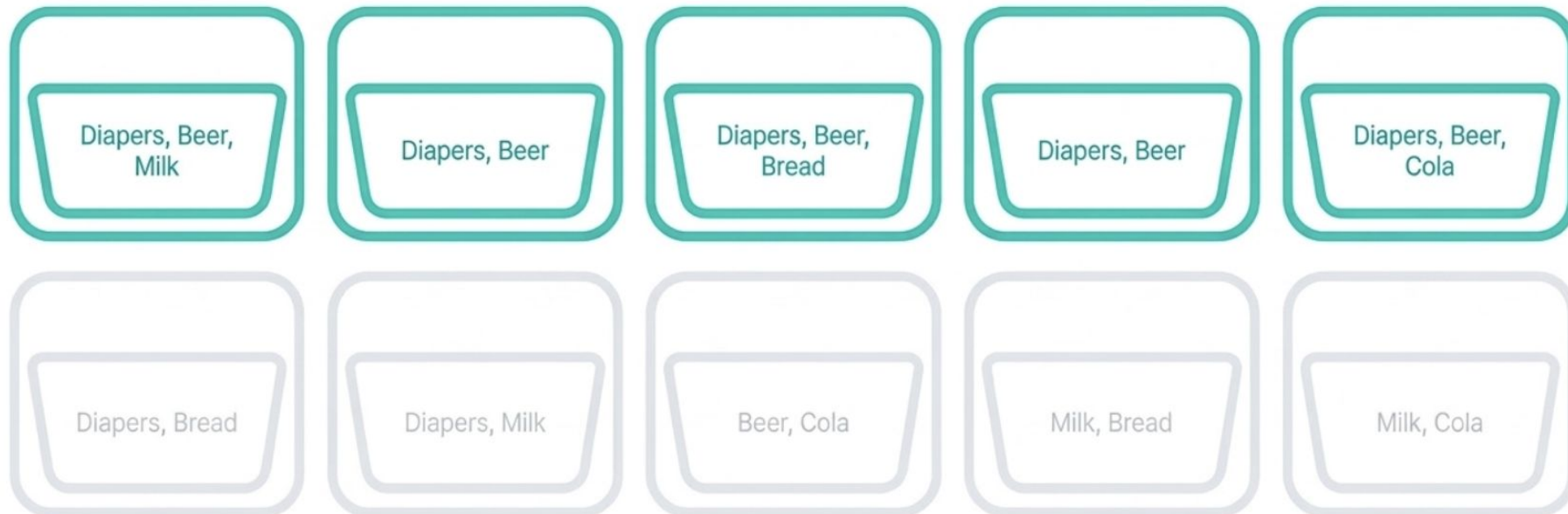
Association Rules

Example

1. Support (How often does this happen?)

5 baskets with [Diaper + Beer] / 10 total baskets = 50%

Support measures popularity. If it's too low, the rule isn't prevalent enough to warrant business action. Half of all our baskets contain this pairing.



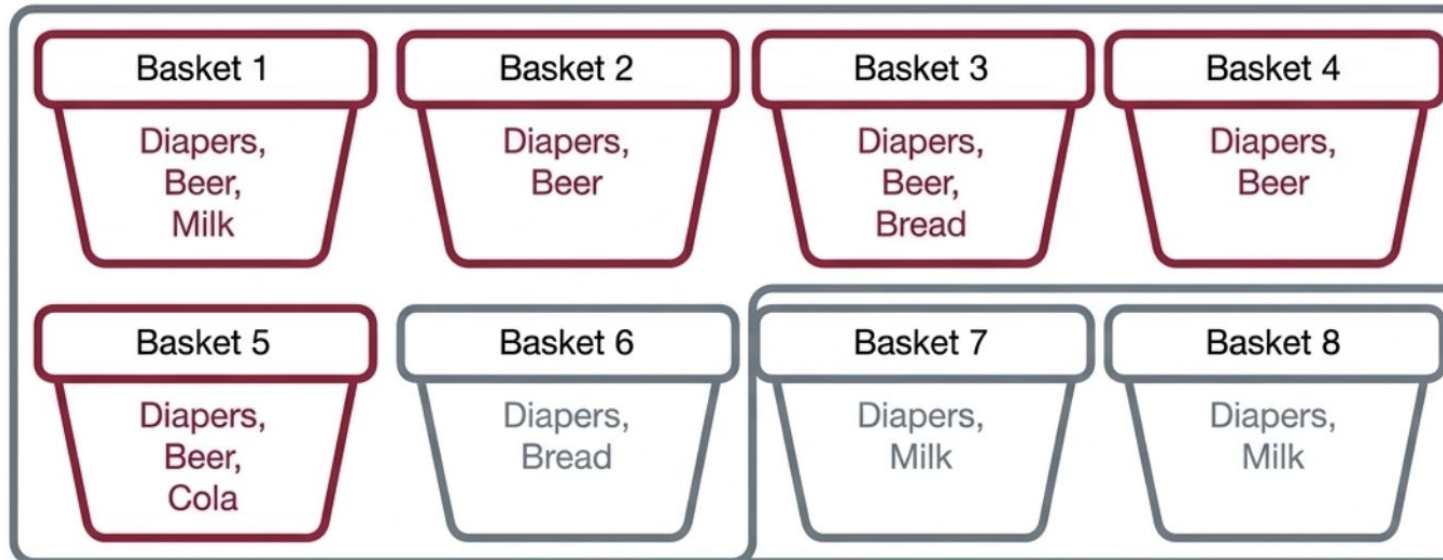
Association Rules

Example

2. Confidence (How reliable is the rule?)

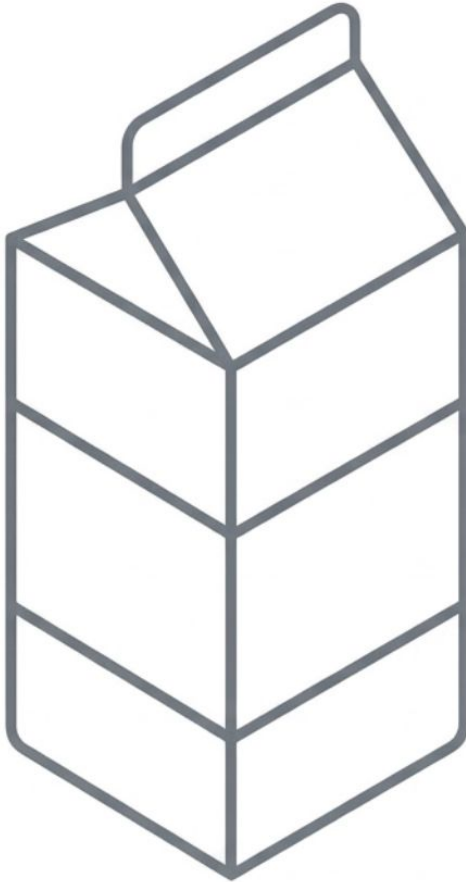
5 baskets with [Beer] / 7 baskets with [Diapers] = 71%

Confidence measures reliability. Out of **everyone who buys diapers**, 71% will reliably buy beer.



Association Rules

Example



The Baseline Illusion

Why isn't a 71% Confidence score enough?

Imagine if Beer was already present in 95% of **all shopping baskets**. In that scenario, a **71% confidence rate** for Diapers → Beer would actually represent a decrease in likelihood.

High confidence can be a **mathematical illusion driven by an item's baseline popularity**.

Association Rules

Example

3. Lift (The Truth Serum)

Insight: Lift compares the rule's confidence against random chance. It **reveals if 'A' actually drives** the purchase of **'B'**, or if they just happened to end up in the same cart.

Confidence of (Diapers → Beer) = 0.71

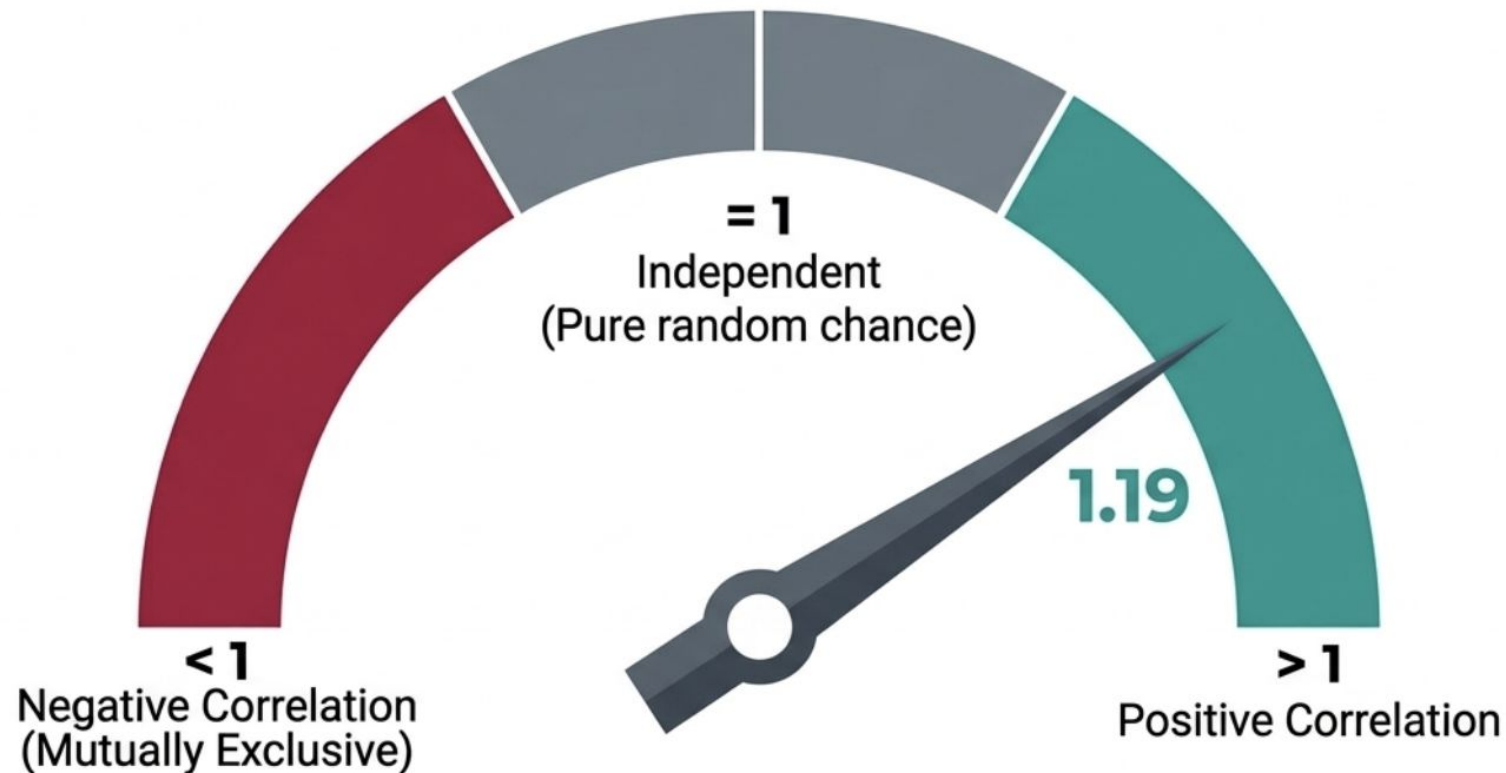
Overall Support of (Beer) = 6/10 = 0.60

= Lift = 1.19

Association Rules

Example

In our example, Lift = 1.19. Because $1.19 > 1$, this is a true positive relationship. Buying diapers actively drives a 19% increase in the likelihood of buying beer.



Association Rules

Example

Direction Matters: $A \rightarrow B \neq B \rightarrow A$

Insight: Support (50%) and Lift (1.19) remain identical in both directions. However, Confidence changes completely because the starting group (the denominator) changes. Placing diapers near the beer aisle (83% confidence) is a stronger move than placing beer near diapers.

Diapers $\rightarrow$ Beer

Base: 7 Diaper buyers,
5 bought beer.

Confidence: $5/7 = 71\%$

Beer $\rightarrow$ Diapers

Base: 6 Beer buyers,
5 bought diapers.

Confidence: $5/6 = 83\%$

Association Rules

Example

The 3-Step Filtering Process

Diapers → Beer	Support: 0.50	Conf: 0.71	Lift: 1.19
Beer → Diapers	Support: 0.50	Conf: 0.83	Lift: 1.19
Milk → Cola	Support: 0.10	Conf: 0.25	Lift: 0.83

Step 1: Discard $\text{Lift} \leq 1$.
Rule Milk → Cola (0.83) is immediately eliminated.

Step 2: Ensure high Support.
The pattern must be frequent enough to matter.

Step 3: Ensure high Confidence.
The pattern must be reliable enough to trust.

Association Rules

Example

- Apply association rules for this table

Home	Insulation	Avg_Age	Num_Occupants	Home_Size	Heating_Oil
H1	3	20	3	4	180
H2	4	68	2	4	230
H3	5	25	5	4	260
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H11	12	47	12	8	560

Association Rules

Example

- Association rules work with items, not continuous numerical values. Therefore, we discretize (bin) each variable into levels, transforming each house into a set of items:

```
Insulation : >= 8      -> Insul=High      ; nguoc lai Insul=Low
Avg_Age     : <35 Young ; 35-52 Middle    ; >52 Old
Num_Occup.  : >= 5      -> Occ=Many        ; nguoc lai Occ=Few
Home_Size   : >= 8      -> Size=Big        ; nguoc lai Size=Std
Heating_Oil : > 360     -> Oil=High (target) ; nguoc lai Oil=Low
```

Association Rules

Example

For the rule $A \rightarrow B$, where N = the total number of transactions:

Support ($A \rightarrow B$) = $\text{count}(A \text{ va } B) / N$ (do pho bien)
Confidence ($A \rightarrow B$) = $\text{count}(A \text{ va } B) / \text{count}(A)$ (do tin cay)
Lift ($A \rightarrow B$) = $\text{Confidence} / [\text{count}(B) / N]$ (so voi ngau nhien; >1 moi dang dung)

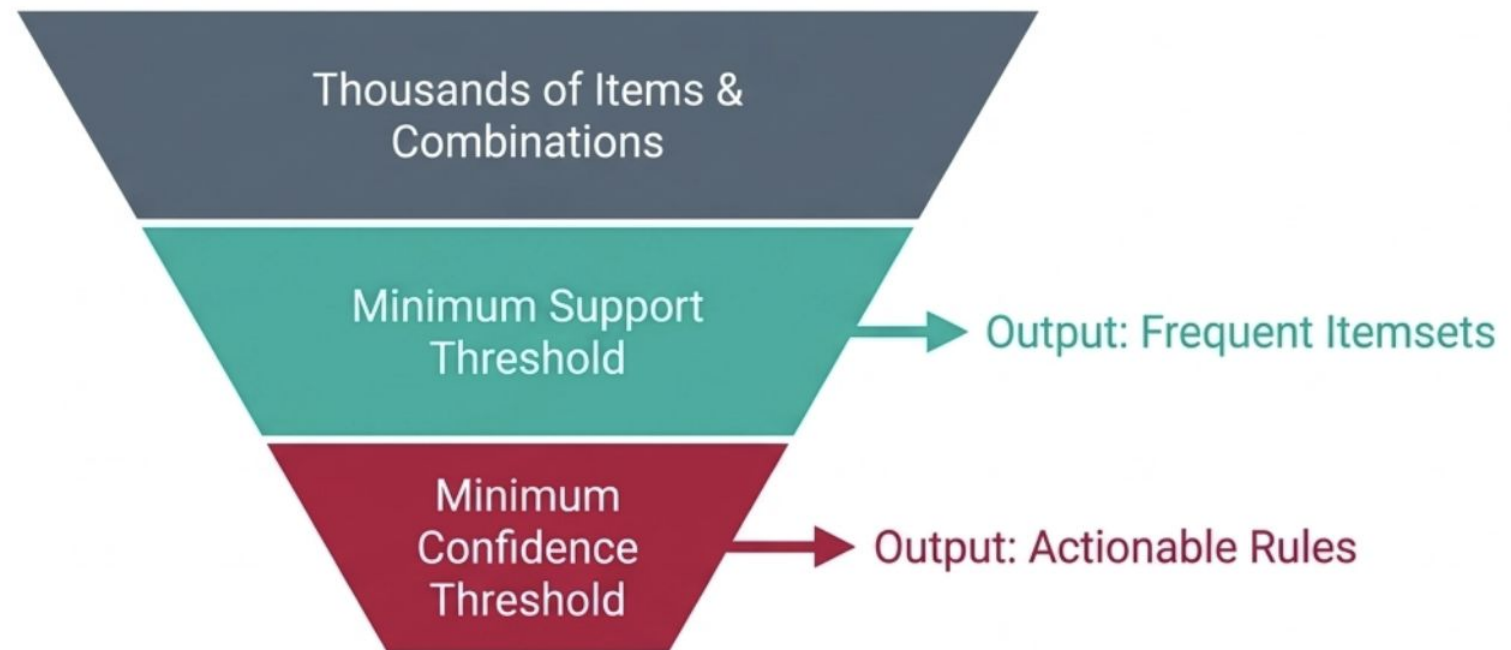
- Complete this table

Luật $A \rightarrow B$	count(A)	count(B)	count(A&B)	Support	Confidence	Lift
Insul=High $\rightarrow$ Oil=High						
Age=Middle $\rightarrow$ Oil=High						
Occ=Many $\rightarrow$ Oil=High						
Size=Big $\rightarrow$ Oil=High						

Association Rules

Summarize

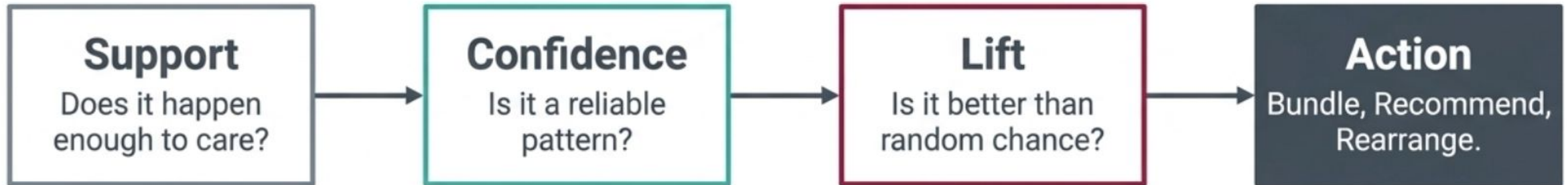
Real stores have thousands of items. Testing every combination is mathematically impossible. Algorithms like Apriori and FP-Growth solve this by generating Frequent Itemsets—they aggressively filter out noise using your Minimum Support threshold before ever calculating rules.



Association Rules

Summarize

The transition from raw transaction data to strategic business action requires all three lenses. **Never rely on one metric** in isolation.



Association Rules

Summarize

Association rules map relationships in any dataset where events occur together.



E-commerce

Frequently bought together engines on Shopee and Amazon.



Retail Strategy

Designing promotional combo-packs and cross-selling.



Healthcare

Analyzing co-occurring symptoms to diagnose syndromes faster.



City Management

Identifying demographic groups that utilize the same community services.

Content

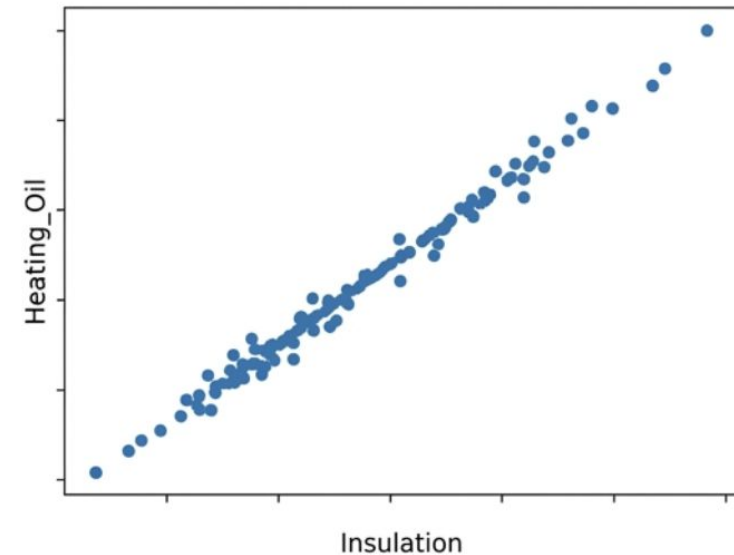
- Correlation
- Association Rules
- **Tools**

Correlation

```
import pandas as pd, matplotlib.pyplot as plt
df = pd.read_csv("heating_oil_dataset.csv")

df.corr(numeric_only=True) # Full correlation matrix
df["Insulation"].corr(df["Heating_Oil"]) # Single pair
df["Insulation"].corr(df["Heating_Oil"]) # Single pair
# Output: r = 0.997

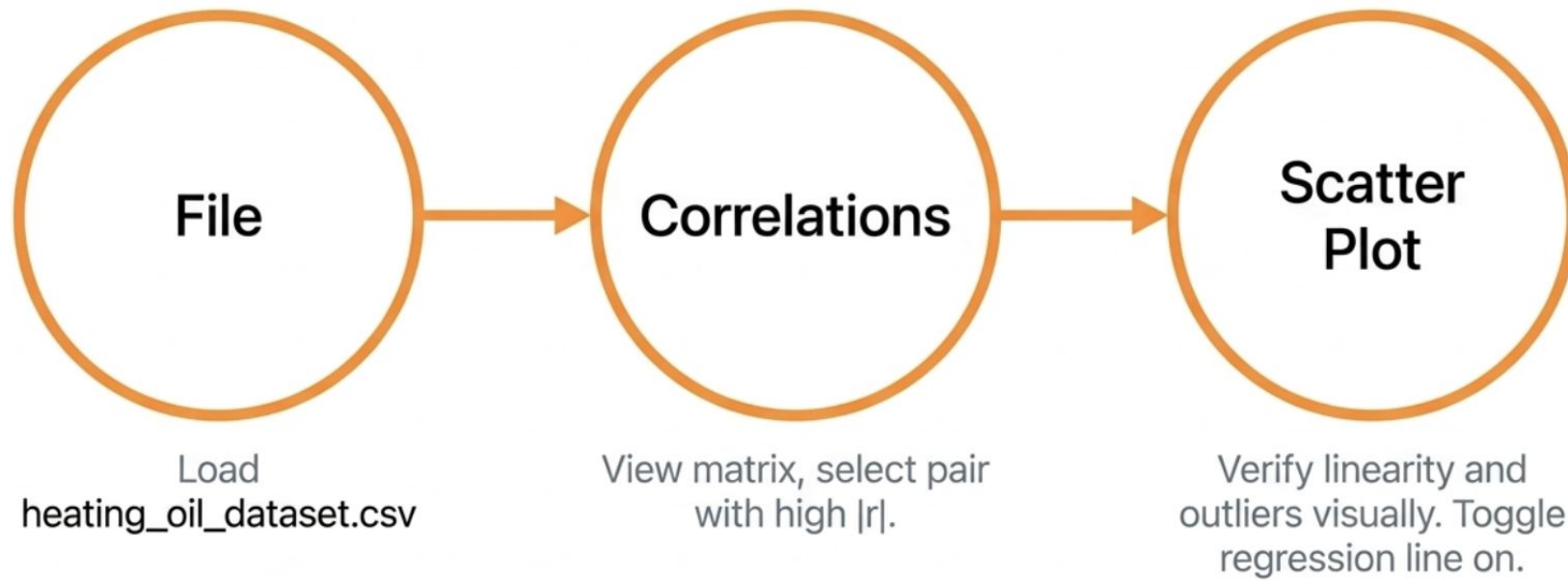
# Visual Validation
plt.scatter(df["Insulation"], df["Heating_Oil"])
plt.xlabel("Insulation"); plt.ylabel("Heating_Oil"); plt.show()
```



The r-value is not enough. Always validate visually with **plt.scatter()** to confirm linearity and spot hidden outliers.

Tools

Correlation



Unlike Python, Correlation is a core, built-in widget in Orange—no external libraries required.

Tools

Association Rules

The Golden Rule: Continuous data must be discretized before mining rules.
(Association rules require categories, not raw numbers).

Method: `pd.cut()` (Binning)

```
df["Oil_lv"] =  
    pd.cut(df["Heating_Oil"],  
           bins=[0,360,999],  
           labels=["Low","High"])  
  
df["Insul_lv"] =  
    pd.cut(df["Insulation"],  
           bins=[0,7,99],  
           labels=["Low","High"])
```

Method: Discretize Widget

Data Input (Insulation, Avg_Age, Heating_Oil)

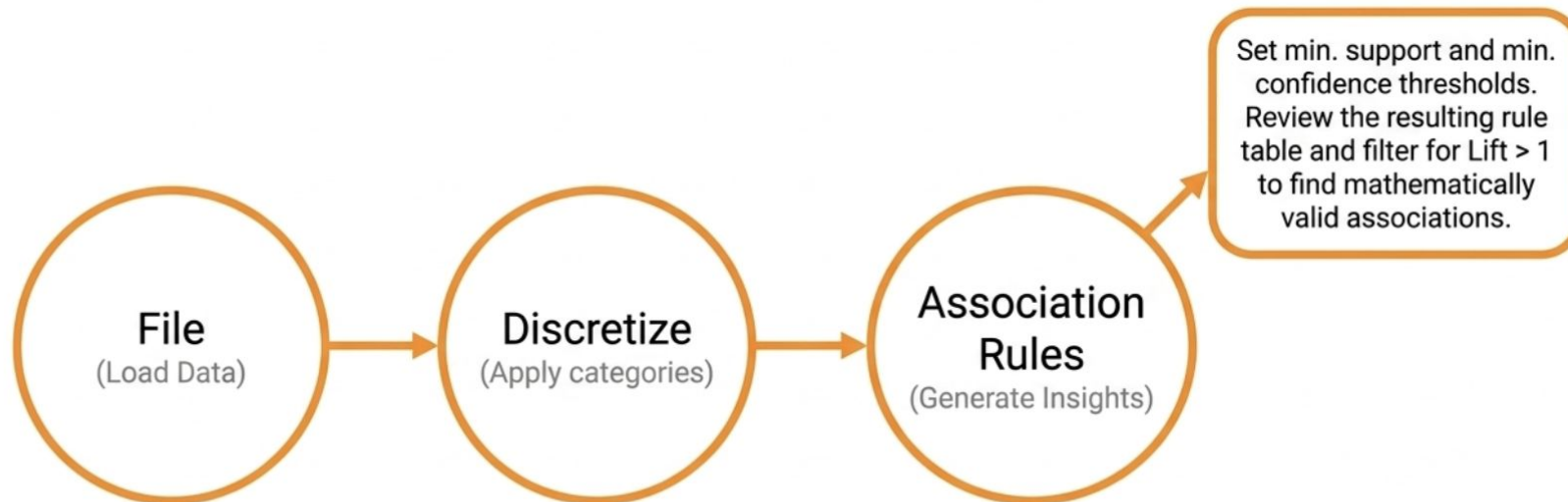
Discretize Widget

[Low] / [Middle] / [High]

Tools

Association Rules

Setup Required: Go to Options → Add-ons → Tick 'Associate' → Restart Orange.



Summarize

Minimalist Documentation

Feature	Python	Orange
Setup for Rules	Needs mlxtend library installed	Needs 'Associate' Add-on installed
Discretization	Code-driven (<code>pd.cut</code> with custom manual bins)	UI-driven (Discretize widget with visual binning)
Rule Generation	Requires building boolean masks or <code>apriori()</code> loops	Instant, interactive tabular output with built-in filtering
Ideal Use Case	Building automated, scalable data pipelines	Rapid exploratory data analysis and presentation

Thank you!